

Climate 5000 Series - Gen 4

18K BTU Single Zone Regular System

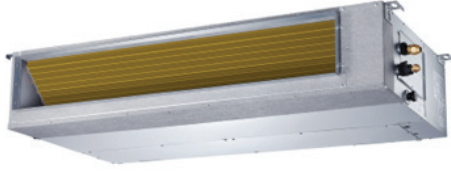
Ducted IDU

Engineering
 Submittal
 Sheet



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Overview and Project Information



Project name	
Location	
Architect	
Engineer	
General contractor	
Reference #	

18K Btu Single Zone Regular System - Ducted IDU

Indoor part number		8-733-962-711
Indoor model number		BMS500-AAU018-1AHDXD
Outdoor part number		8-733962-686
Outdoor model number		BMS500-AAS018-1CSXRD
Efficiency Data		AHRI 210/240 2023
SEER2		20.0
EER2		12.5
HSPF2 Region IV		10.5
HSPF2 Region V		8.2
COP(47F)		3.47
COP (5F)		2.20
Performance Data		AHRI 210/240 2023
Cooling	Capacity Range	7000~19000
	Rated Capacity	17000
	Current	6.2
	Input Power	1445
Heating	Capacity Range	5500~22000
	Rated Capacity (47F)	20000
	Rated Capacity (5F)	14000
	Current	7.2
	Input Power	1690
Electrical Data		
Power supply (V / Ø / Hz)		208-230 / 1 / 60
Min. circuit ampacity indoor/outdoor (A)		3.0 / 18.0
Max. fuse indoor/outdoor (A)		15.0 / 20.0
Compressor Data		
Type:		Rotary
Refrigerant oil /oil charge (ml)		VG74 / 440
Input (W)		1090
Rated current (RLA/A)		7.2
Piping		
Refrigerant precharge (oz/kg)		47.97 / 1.36
Length for precharge (ft/m)		24.6 / 7.5
Liquid side (in/mm)		1/4" / Ø6.35
Gas side (in/mm)		1/2" / Ø12.7
Additional Charge of Refrigerant (oz/ft)/(g/m)		0.16 / 15
Max. refrigerant pipe length (ft/m)		98.40 / 30
Max. difference in height (ft/m)		65.60 / 20

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Operation Range

ODU	Cooling (°F / °C)	-13 ~ 122 / -25 ~ 50
	Heating (°F / °C)	-13 ~ 75 / -25 ~ 24
IDU	Cooling (°F / °C)	60 ~ 90 / 16 ~ 32
	Heating (°F / °C)	32 ~ 86 / 0 ~ 30

System Data

Refrigerant type		R454B
Design pressure (PSIG)		550 / 340
Dehumidification (pts/hr)		3.22
IDU	Fan motor model	ZKFN-165-10-1L
	Fan motor qty (ea)	1
	Fan motor input (W)	185.0
	Fan speed, H / M / L (RPM)	1100 / 900 / 700
	Air flow, Turbo / H / M / L / Si (CFM)	600.4 / 559.2 / 500.3 / 441.4 / N/A
	Noise level, Turbo / H / M / L / Si (dB(A))	N/A / 40.5 / 39 / 38 / 35
	Net weight (lbs / kg)	54.45 / 24.7
	Gross weight (lbs / kg)	66.14 / 30.0
ODU	ESP range (Pa)	0-100
	Drain pipe outside diameter in. (mm)	OD 0.98 (25)
	Fan motor model	ZKFN-80-10-1L
	Fan motor qty (ea)	1
	Fan motor input (W)	161
	Fan motor speed, H / / L (RPM)	780 / 700 / 550
	Air flow, Max (CFM)	1765.8
	Noise level, Max (dB(A))	59.0
Accessories	Net weight (lbs / kg)	92.59 / 42.0
	Gross weight (lbs / kg)	100.31 / 45.5

Accessories

BMS-WT2-XXD (8733962707)	Wired Wall Thermostat
BMS-DRY-XXD (8733969263)	24V Interface
BMS-PK2-7X1 (8733951011)	25ft Lineset

Bosch Thermotechnology Corp.
 Londonderry, NH • Watertown, MA • Ft. Lauderdale, FL

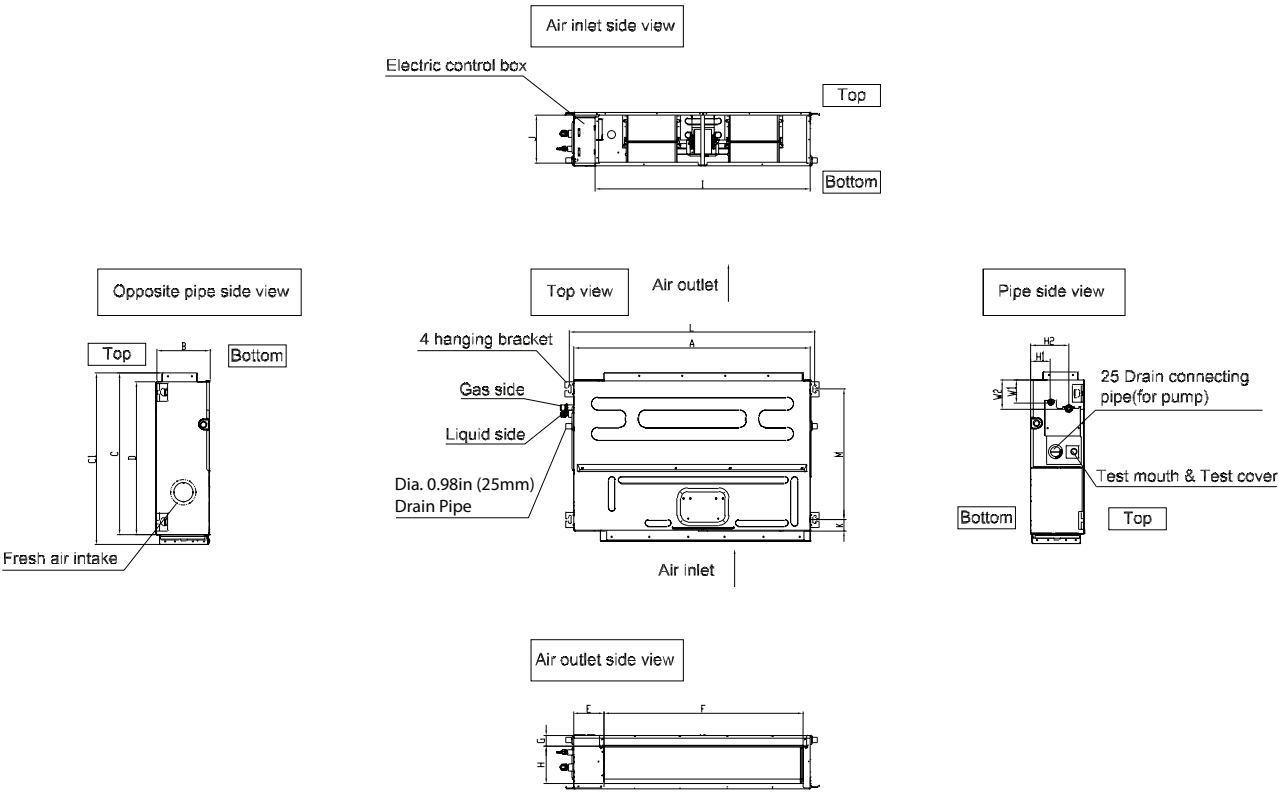
Tel: 1-800-283-3787 www.bosch-homecomfort.us

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Indoor Unit Dimensions and Connections

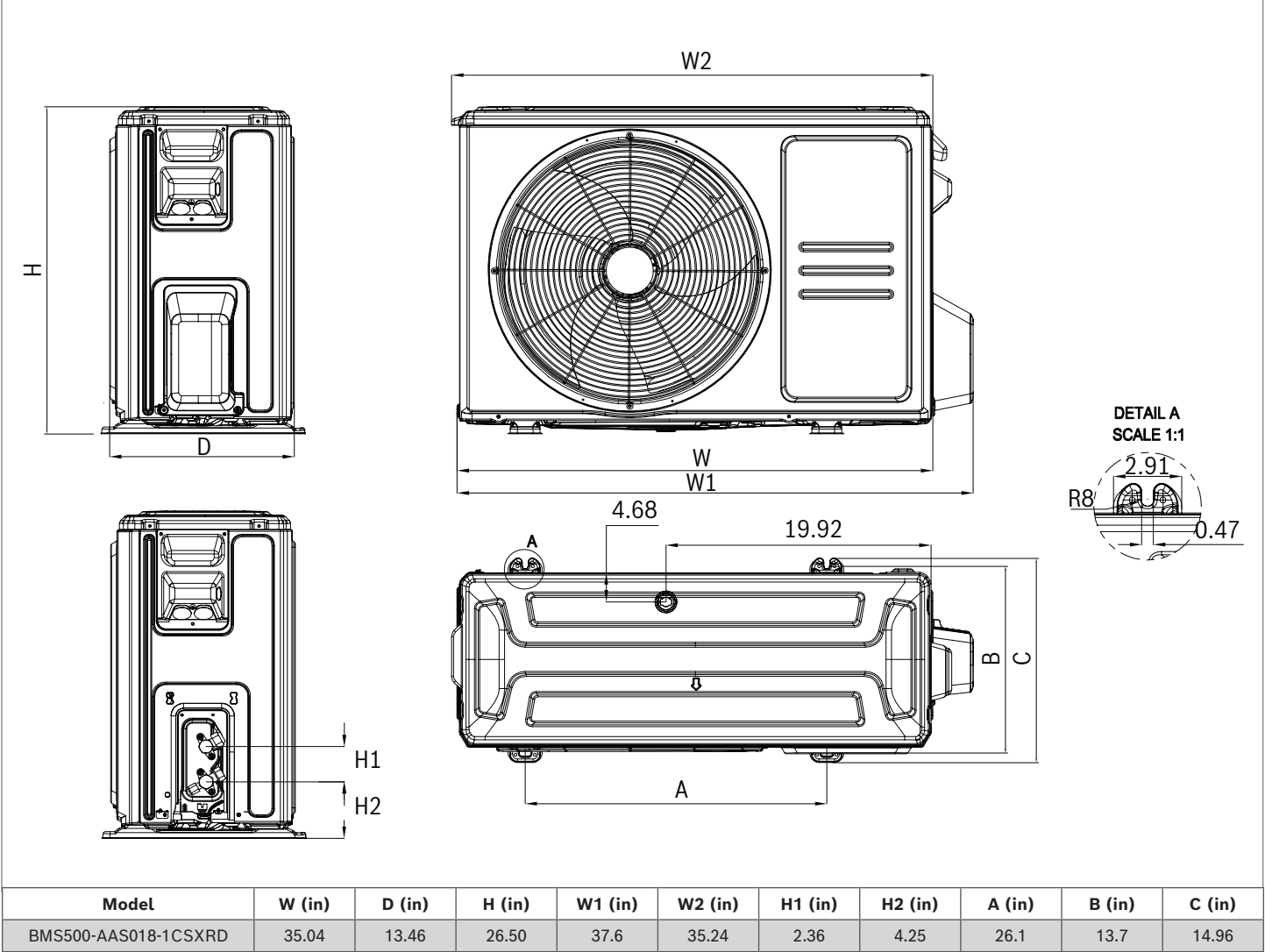


Model	Outline dimension (in)				Air outlet dimension (in)				Return air (in)			Mounting (in)		Gas and liquid valve (in)			
	A	B	C	D	E	F	G	H	I	J	K	L	M	H1	H2	W1	W2
BMS500-AAU018-1AHDXD	34.7	8.3	26.5	23.6	5.5	27.8	2.0	5.4	30.8	7.5	1.6	36.2	20.0	3.1	5.8	3.5	4.4

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Outdoor Unit Dimensions and Connections



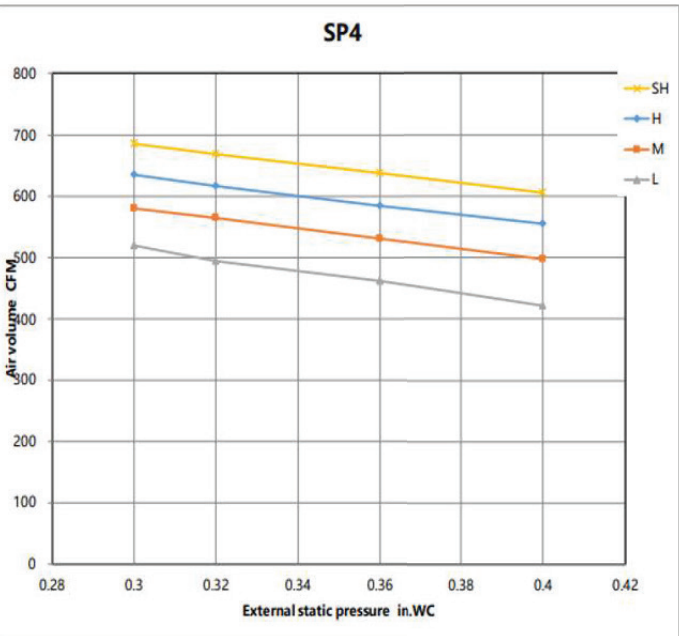
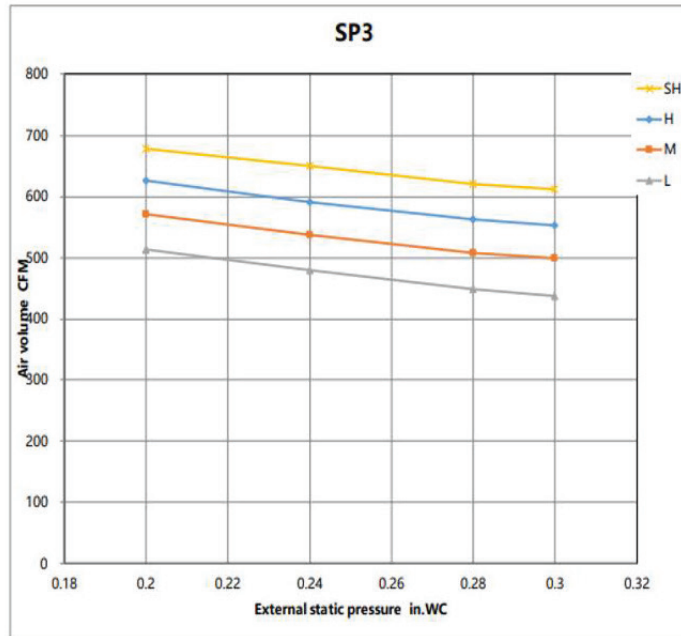
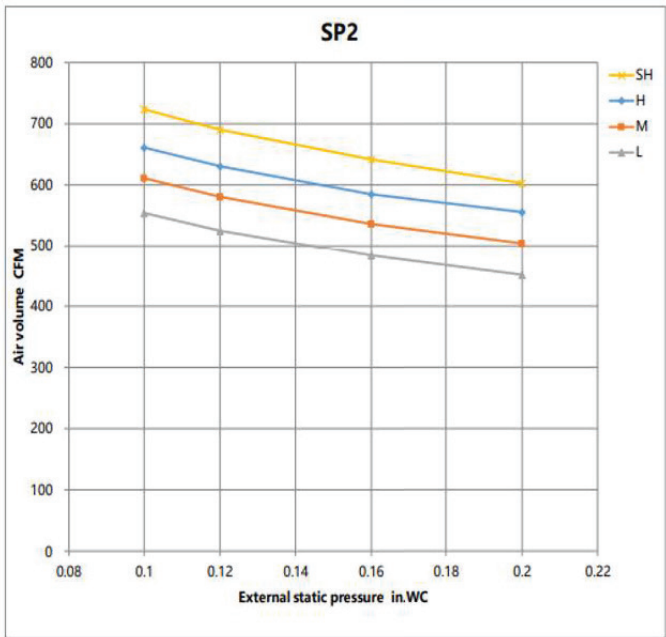
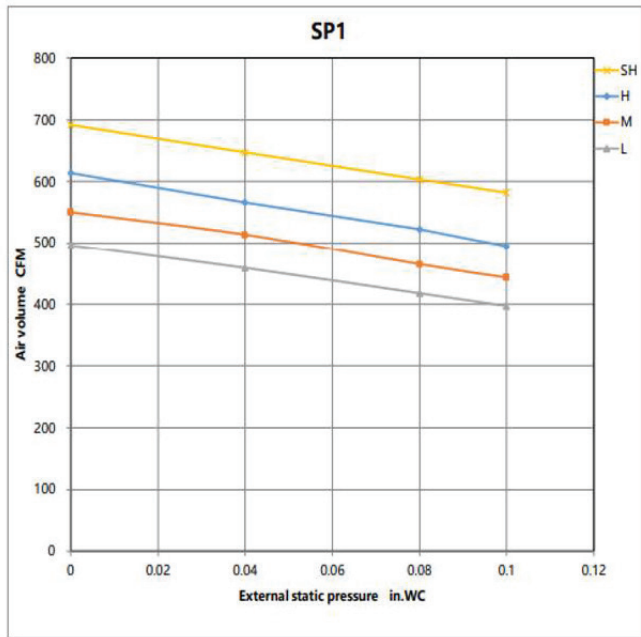
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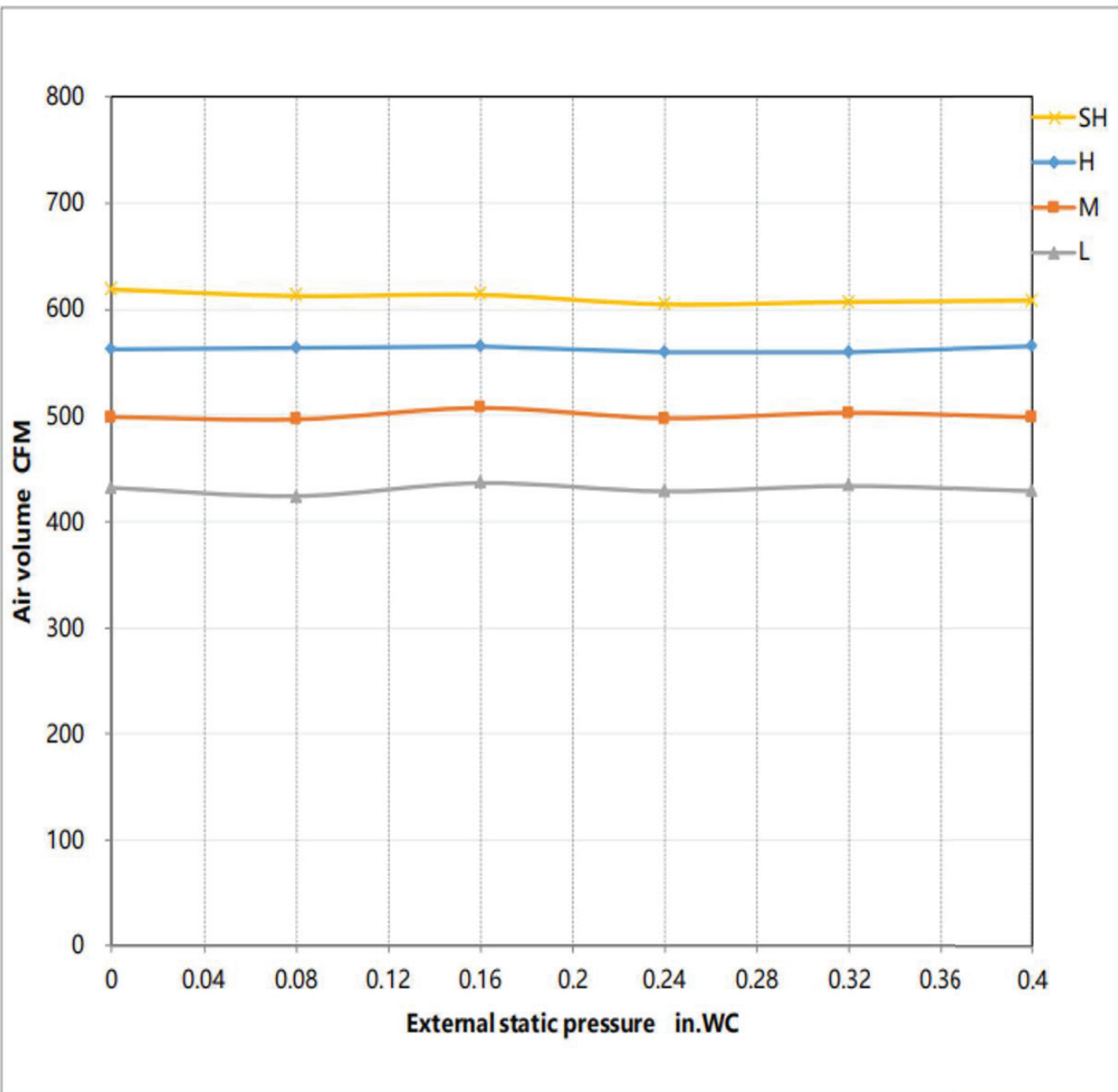


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Air Volume



Air Volume



Climate 5000 Series - Gen 4

18K Btu Single Zone Regular System - Ducted IDU


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Cooling Performance - Indoor Dry Bulb Temperature at 600 CFM																					
ODU DB(°F)	IDU WB (°F)	60.8°F(16.0°C)					64.4°F(18.0°C)					67°F(19.4°C)					71.6°F(22.0°C)				
	IDU DB (°F)	73.4°F (23.0°C)	77.0°F (25.0°C)	80.0°F (26.7°C)	80.6°F (27.0°C)	84.2°F (29.0°C)	73.4°F (23.0°C)	77.0°F (25.0°C)	80.0°F (26.7°C)	80.6°F (27.0°C)	84.2°F (29.0°C)	73.4°F (23.0°C)	77.0°F (25.0°C)	80.0°F (26.7°C)	80.6°F (27.0°C)	84.2°F (29.0°C)	73.4°F (23.0°C)	77.0°F (25.0°C)	80.0°F (26.7°C)	80.6°F (27.0°C)	84.2°F (29.0°C)
-13°F (-25.0°C)	TC	16046	16039	16267	16335	16784	16742	16728	16717	16716	16702	17242	17221	17204	17202	17185	18187	18182	18145	18138	18100
	S/T	0.80	0.91	1.00	1.00	1.00	0.65	0.75	0.85	0.86	0.97	0.55	0.65	0.74	0.76	0.86	0.39	0.47	0.56	0.57	0.67
	PI	0.58	0.58	0.58	0.58	0.58	0.58	0.58	0.58	0.58	0.58	0.58	0.58	0.58	0.58	0.58	0.58	0.58	0.58	0.58	0.58
-4°F (-20.0°C)	TC	17280	17272	17519	17591	18076	18029	18015	18003	18002	17987	18569	18545	18528	18525	18507	19586	19581	19541	19533	19492
	S/T	0.80	0.92	1.00	1.00	1.00	0.65	0.76	0.85	0.87	0.98	0.55	0.65	0.74	0.76	0.86	0.39	0.48	0.56	0.57	0.67
	PI	0.60	0.61	0.60	0.60	0.60	0.60	0.60	0.60	0.60	0.60	0.60	0.60	0.60	0.60	0.60	0.60	0.60	0.60	0.60	0.60
0°F (-17.8°C)	TC	17854	17843	18174	18250	18751	18620	18599	18585	18582	18769	19167	19140	19116	19113	19104	20198	20193	20191	20194	20207
	S/T	0.81	0.93	1.00	1.00	1.00	0.66	0.77	0.86	0.88	1.00	0.55	0.66	0.75	0.77	0.87	0.39	0.48	0.57	0.58	0.68
	PI	0.61	0.61	0.61	0.61	0.61	0.61	0.61	0.61	0.61	0.61	0.61	0.61	0.61	0.61	0.61	0.62	0.62	0.62	0.62	0.62
5°F (-15.0°C)	TC	18366	18429	18850	18937	19521	19227	19232	19233	19253	19540	19842	19847	19851	19852	19878	21006	20999	21010	21013	21022
	S/T	0.82	0.94	1.00	1.00	1.00	0.66	0.78	0.87	0.88	1.00	0.56	0.67	0.76	0.78	0.88	0.39	0.48	0.57	0.59	0.69
	PI	0.61	0.61	0.62	0.62	0.62	0.62	0.62	0.62	0.62	0.62	0.62	0.62	0.62	0.62	0.62	0.64	0.64	0.64	0.64	0.64
14°F (-10.0°C)	TC	18521	18512	18889	18978	19567	19440	19423	19410	19408	19590	20106	20081	20061	20058	20037	21380	21373	21332	21325	21279
	S/T	0.81	0.93	1.00	1.00	1.00	0.65	0.76	0.86	0.87	1.00	0.55	0.66	0.75	0.76	0.87	0.39	0.48	0.56	0.57	0.67
	PI	0.69	0.69	0.69	0.69	0.69	0.69	0.69	0.69	0.69	0.69	0.69	0.69	0.69	0.69	0.69	0.69	0.69	0.69	0.69	0.69
17°F (-8.3°C)	TC	18529	18590	19019	19112	19733	19451	19459	19463	19479	19754	20115	20123	20131	20132	20157	21393	21387	21384	21388	21406
	S/T	0.82	0.93	1.00	1.00	1.00	0.66	0.77	0.87	0.88	1.00	0.55	0.66	0.76	0.77	0.87	0.39	0.48	0.57	0.58	0.68
	PI	0.71	0.71	0.71	0.71	0.71	0.71	0.71	0.71	0.71	0.71	0.71	0.71	0.71	0.71	0.71	0.72	0.72	0.72	0.72	0.72
23°F (-5.0°C)	TC	18304	18382	18858	18951	19575	19230	19236	19238	19263	19596	19894	19900	19906	19906	19938	21153	21146	21161	21164	21176
	S/T	0.83	0.95	1.00	1.00	1.00	0.66	0.78	0.88	0.89	1.00	0.56	0.67	0.76	0.78	0.89	0.39	0.48	0.57	0.59	0.69
	PI	0.73	0.73	0.73	0.73	0.74	0.73	0.73	0.73	0.73	0.74	0.74	0.74	0.74	0.74	0.74	0.75	0.75	0.75	0.75	0.75
35°F (1.7°C)	TC	18023	18023	18129	18180	18766	18865	18864	18866	18866	18916	19456	19454	19455	19456	19458	20551	20546	20545	20544	20546
	S/T	0.78	0.88	0.97	0.99	1.00	0.63	0.73	0.82	0.83	0.93	0.54	0.63	0.72	0.73	0.83	0.39	0.46	0.54	0.56	0.65
	PI	0.72	0.72	0.72	0.72	0.75	0.76	0.76	0.76	0.76	0.76	0.79	0.79	0.79	0.79	0.79	0.86	0.86	0.86	0.86	0.86
41°F (5.0°C)	TC	17611	17611	17751	17875	18403	18430	18430	18431	18431	18495	19005	19004	19005	19005	19006	20071	20066	20065	20064	20066
	S/T	0.78	0.89	0.98	1.00	1.00	0.64	0.74	0.83	0.84	0.94	0.54	0.64	0.72	0.74	0.84	0.39	0.47	0.55	0.56	0.66
	PI	0.74	0.74	0.75	0.76	0.79	0.79	0.79	0.79	0.79	0.79	0.83	0.83	0.83	0.83	0.83	0.90	0.90	0.90	0.90	0.90
47°F (8.3°C)	TC	17199	17210	17441	17519	18037	17996	17995	17996	17996	18079	18555	18553	18554	18554	18554	19594	19589	19587	19586	19588
	S/T	0.79	0.90	1.00	1.00	1.00	0.64	0.75	0.84	0.85	0.96	0.54	0.65	0.73	0.75	0.85	0.39	0.47	0.55	0.57	0.67
	PI	0.77	0.77	0.79	0.79	0.83	0.82	0.82	0.82	0.82	0.83	0.86	0.86	0.86	0.86	0.86	0.94	0.94	0.94	0.94	0.94
50°F (10.0°C)	TC	16913	17379	17965	18069	18770	17868	17878	18001	18058	18787	18565	18589	18599	18600	18779	19844	19897	19948	19954	19979
	S/T	0.90	1.00	1.00	1.00	1.00	0.71	0.85	0.96	0.99	1.00	0.59	0.72	0.83	0.85	0.99	0.40	0.50	0.60	0.62	0.74
	PI	0.88	0.88	0.88	0.88	0.88	0.88	0.88	0.88	0.88	0.88	0.88	0.88	0.88	0.88	0.88	0.88	0.88	0.88	0.88	0.88
59°F (15.0°C)	TC	16979	17575	18199	18310	19055	17965	17989	18194	18329	19073	18706	18725	18731	18759	19089	20087	20127	20167	20171	20188
	S/T	0.92	1.00	1.00	1.00	1.00	0.73	0.86	1.00	1.00	1.00	0.60	0.73	0.85	0.86	1.00	0.40	0.51	0.61	0.63	0.76
	PI	0.93	0.93	0.94	0.94	0.94	0.93	0.93	0.94	0.94	0.94	0.94	0.94	0.94	0.94	0.94	0.94	0.94	0.94	0.94	0.94
68°F (20.0°C)	TC	17037	17764	18423	18540	19327	18036	18077	18443	18560	19346	18816	18830	18864	18881	19362	20275	20312	20344	20347	20358
	S/T	0.95	1.00	1.00	1.00	1.00	0.74	0.89	1.00	1.00	1.00	0.61	0.75	0.87	0.89	1.00	0.40	0.51	0.63	0.65	0.78
	PI	0.98	0.99	0.99	0.99	1.00	0.99	0.99	0.99	0.99	1.00	1.00	1.00	1.00	1.00	1.00	1.01	1.01	1.01	1.01	1.01
77°F (25.0°C)	TC	17080	17903	18603	18727	19564	18107	18165	18624	18748	19584	18927	18949	18997	19018	19600	20442	20504	20552	20557	20576
	S/T	0.97	1.00	1.00	1.00	1.00	0.75	0.90	1.00	1.00	1.00	0.61	0.76	0.88	0.90	1.00	0.40	0.52	0.63	0.65	0.79
	PI	1.04	1.05	1.05	1.05	1.06	1.05	1.05	1.05	1.05	1.06	1.05	1.05	1.05	1.05	1.06	1.07	1.06	1.06	1.06	1.06
82°F (27.8°C)	TC	17113	17986	18712	18839	19701	18134	18204	18728	18860	19722	18978	19002	19061	19084	19739	20519	20596	20650	20655	20678
	S/T	0.99	1.00	1.00	1.00	1.00	0.76	0.91	1.00	1.00	1.00	0.62	0.77	0.89	0.92	1.00	0.40	0.52	0.64	0.66	0.80
	PI	1.15	1.15	1.16	1.16	1.16	1.15	1.15	1.15	1.16	1.16	1.16	1.16	1.16	1.16	1.16	1.17	1.17	1.17	1.17	1.17
86°F (30.0°C)	TC	17208	18067	18806	18937	19819	18148	18231	18827	18959	19840	19007	19029	19099	19127	19857	20578	20657	20706	20711	20731
	S/T	1.00	1.00	1.00	1.00	1.00	0.77	0.93	1.00	1.00	1.00	0.62	0.78	0.91	0.93	1.00	0.40	0.52	0.64	0.66	0.81
	PI	1.32	1.33	1.34	1.34	1.34	1.33	1.33	1.33	1.34	1.34	1.34	1.34	1.34	1.34	1.34	1.36	1.35	1.35	1.35	1.35
95°F (35.0°C)	TC	17295	17388	17950	18057	18771	18381	18392	18398	18419	18794	19170	19174	19186	19187	19219	20656	20649	20653	20659	20686
	S/T	0.83	0.95	1.00	1.00	1.00	0.66	0.78	0.88	0.89	1.00	0.56	0.67	0.76	0.78	0.88	0.39	0.48	0.57	0.58	0.68
	PI	1.65	1.66	1.68	1.69	1.72	1.70	1.70	1.70	1.71	1.72	1.74	1.74	1.74	1.74	1.75	1.81	1.81	1.81	1.81	1.81
104°F (40.0°C)	TC	14547	14680	15233	15325	15936	15464	15472	15500	15516	15955	16123	16130	16138	16139	16185	1				

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Heating Performance - Indoor Dry Bulb Temperature at 600 CFM										
Outdoor DB(°F)	TC Total Heating Capacity(BTU/hr)					PI Power Input(kW)				
	Indoor Conditions(DB °F)					Indoor Conditions(DB °F)				
	60.8°F (16.0°C)	68°F (20.0°C)	70°F (21.1°C)	71.6°F (22.0°C)	75.2°F (24.0°C)	60.8°F (16.0°C)	68°F (20.0°C)	70°F (21.1°C)	71.6°F (22.0°C)	75.2°F (24.0°C)
-13°F(-25.0°C)	9497	9307	9260	9223	9140	1.46	1.47	1.47	1.47	1.48
-4°F(-20.0°C)	10560	10349	10297	10256	10164	1.47	1.48	1.48	1.48	1.49
0°F(-17.8°C)	12252	12007	11947	11900	11792	1.56	1.57	1.57	1.57	1.58
5°F(-15.0°C)	14367	14080	14010	13954	13828	1.67	1.68	1.69	1.69	1.70
14°F(-10.0°C)	14730	14435	14363	14305	14177	1.67	1.68	1.69	1.69	1.70
17°F(-8.3°C)	14850	14553	14481	14423	14293	1.67	1.69	1.69	1.69	1.70
22°F(-5.6°C)	15779	15463	15386	15324	15186	1.69	1.71	1.71	1.71	1.72
27°F(-2.8°C)	16707	16373	16291	16226	16080	1.72	1.73	1.73	1.74	1.74
32°F(0.0°C)	17635	17282	17196	17127	16973	1.74	1.75	1.75	1.76	1.77
35°F(1.7°C)	18192	17828	17739	17668	17509	1.75	1.76	1.77	1.77	1.78
37°F(2.8°C)	19008	18628	18535	18460	18294	1.76	1.77	1.77	1.78	1.79
42°F(5.6°C)	21048	20627	20523	20441	20257	1.77	1.79	1.79	1.80	1.80
44.6°F(7.0°C)	21863	21426	21319	21234	21043	1.78	1.80	1.80	1.80	1.81
47°F(8.3°C)	23087	22626	22512	22422	22221	1.79	1.81	1.81	1.81	1.82
52°F(11.1°C)	22348	21901	21792	21705	21509	1.62	1.63	1.63	1.63	1.64
57°F(13.9°C)	21609	21177	21071	20987	20798	1.44	1.45	1.45	1.46	1.46
59°F(15.0°C)	21314	20887	20783	20700	20513	1.37	1.38	1.38	1.39	1.39
62°F(16.7°C)	20870	20453	20350	20269	20087	1.26	1.27	1.28	1.28	1.28
64.4°F(18.0°C)	20515	20105	20005	19925	19745	1.18	1.19	1.19	1.19	1.20
69.8°F(21.0°C)	15564	15253	15177	15116	14980	0.99	1.00	1.00	1.00	1.01
75.2°F(24.0°C)	14147	13864	13795	13740	13616	0.80	0.81	0.81	0.81	0.81

Bolded values were tested. All other values are simulated and for reference only.